Deep Learning-Based Plant Leaf Disease Detection Using Transfer Learning and Ensemble Models

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Abstract: **A complete method for finding diseases on plant leaves is described in this study. It uses advanced deep learning techniques, especially transfer learning and ensemble models. The study uses the PlantVillage Dataset to its full potential. To do this, it first normalises the images and adds more data using techniques like flipping, rotating, changing the saturation, scaling, translating, and shearing. The images are then improved by turning them into greyscale, using Gaussian blur, and looking for edges. MobileNetV2 and DenseNet121, the two main deep learning models, were tweaked to work best for identifying plant diseases. It was even more innovative to combine these models into a small ensemble framework that uses the best parts of both designs to make recognition more accurate. The results show that the ensemble model does better than the individual models, with scores of 97% for accuracy, 96% for precision, 96% for recall, and 97% for F1-score. This shows that it is very good at accurately identifying different plant diseases. This study shows that using lightweight models in settings with limited resources is a good idea.**

Keywords: Deep Learning, Plant Diseases, Transfer Learning, Ensemble Models, Image Processing, Precision Agriculture.

I. Introduction

Agricultural technology that helps find plant diseases early is very important for making sure crops are healthy and produce a lot. Deep learning approaches have brought about a significant shift towards more accurate automated illness diagnosis systems [1]. This paper aims to rapidly identify illnesses on plant leaves using deep learning models—especially transfer learning and ensemble approaches. Not only difficult but also prone to errors are conventional methods of illness diagnosis, which largely rely on hand-held object inspection under professional assistance. These issues make it abundantly evident that we need more robust, more automated solutions scalable in nature. One fascinating approach to address these issues might be to assist in the identification of plant diseases using deep learning [2]. Visual data handling ability of pre-trained models such as MobileNetV2 and DenseNet121 is well-known. This study alters these designs to improve their ability to detect illnesses on plant leaves. Comparatively to what is often required to train deep neural networks from scratch, transfer learning allows these models be fine-tuned with a little quantity of input [3], [4]. This makes it a helpful instrument in cases lacking a lot of tagged data.

Furthermore included in this study is a new ensemble model combining the finest aspects of the modified MobileNetV2 and DenseNet121 models. It is well known that ensemble techniques provide more accurate forecasts. They reduce bias and mistake by integrating the qualities of various models, therefore offering a robust framework. This approach significantly helps to increase the reliability and accuracy of illness discovery. This work prepares the data for the optimal learning performance by means of tight pre-processing and data addition techniques using the PlantVillage Dataset. These techniques are supposed to provide data more in line with reality. This prepares the models to handle a spectrum of complicated illness signs [5]. We demonstrate in this work that sophisticated deep learning methods might alter the detection of plant diseases. These approaches provide a scalable and efficient solution that can significantly enhance agricultural practices all throughout the world.

II. Related Work

Deep learning is increasingly used in agriculture, particularly to identify plant diseases, in recent years thanks to several developments. Many various research on many facets of this topic have been conducted, and every one of them has advanced this field in unique manner. Most of the work at initially was done using conventional machine learning techniques, which worked somewhat but struggled with the vast volumes of data required for training and the extremely varied visual circumstances [6], [7]. As the discipline developed, deep learning models especially convolutional neural networks made a significant impact on both accuracy and speed. Among the most significant contributions is transfer learning. When there is little tagged data a typical occurrence in agricultural applications it has been extremely beneficial [8]. Research employing pre-trained networks such as AlexNet and GoogleNet altered to identify plant illnesses revealed that they were much more adept than conventional image processing techniques. Models that have previously been trained on large datasets like ImageNet should be used instead of starting from

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scratch, which takes a lot of time and money, as these studies highlight [9], [10].

Additionally, ensemble models have not been used as much in finding plant diseases, but they show a lot of potential. Ensemble methods help improve predictions by combining results from several models [11]. This lowers the chance of overfitting and makes it easier for the models to work with new data they haven't seen before. When you have to deploy robust models like MobileNet and DenseNet in locations with restricted resources, such farms [13], the notion [12] of combining lightweight versions of them makes sense. Building on past research, this work uses novel ensemble techniques and transfer learning to further the use of compact, efficient models. It does this by redefining accuracy and utility in the realm of plant leaf disease detection [14]. These developments not only enable more sophisticated, scalable agricultural technologies but also help us understand more fully concepts.

III. PROPOSED METHODOLOGY

A. Load Dataset

a. Plant Village Dataset

Popular collection of images of plant leaves, the Plant Village dataset facilitates research and development of machine learning models capable of identifying plant illnesses. With more than 54,000 images of 14 varieties of crops apples, tomatoes, potatoes, and more each designated with a particular disease early blight, late blight, or rust it boasts Groups of the images are based on the kind of plant and the illness that affricates it. Different climatic conditions and varying degrees of leaf damage abound. Deep learning models such as convolutional neural networks (CNNs) to automatically categorise illnesses may find a useful source in this dataset. Its large scale and variety make it ideal for use in transfer learning and ensemble models for controlling plant health.

B. Dataset Pre-processing

a. Image Enhancement

Image enhancement techniques improve photographs so that machine learning models may more readily identify characteristics. Regarding disease detection on plant leaves, approaches of improvement include histogram equalising, brightness and sharpness adjustment, these methods enable you to see crucial elements such leaf structure and disease patterns that would be difficult otherwise.

b. Grayscale conversion

Greyscale conversion simplifies image data by converting colourful images with many channels into one intensity channel. When it comes to identifying plant diseases, where colour information may not always be very useful for categorisation, this approach is rather beneficial.

c. Gaussian Blur:

In an image, a convolution technique using a Gaussian kernel eliminates high-frequency noise and features. We label this Gaussian blur. This approach is particularly useful for identifying illnesses on plant leaves as it eliminates background noise and minor information irrelevant to the model's focus on more prominent elements. Blurring the image helps to minimize little mistakes that might compromise the sorting mechanism.

- Gaussian Kernel Definition

$$\boldsymbol{G(x,y) = \left(\frac{1}{2\pi\sigma^2}\right) * exp\left(-\frac{(x^2 + y^2)}{2\sigma^2}\right)}$$

where (x, y) are the coordinates, and σ is the standard deviation.

- 2. Convolution Operation

$$\boldsymbol{I_{blurred(x,y)} = \Sigma[i = -k\ to\ k]\Sigma[j = -k\ to\ k]I(x+i, y+j) * G(i,j)}$$

where I(x, y) is the original image and G(i, j) is the Gaussian kernel.

- 3. Resulting Blurred Image

$$\boldsymbol{I_{blurred} = I * G}$$

where "*" denotes the convolution operation between the image and the Gaussian kernel.

d. Edge Detection:

Techniques for edge recognition, like the Sobel or Canny methods, are used to find the edges of things in a picture. Edge detection brings out the shapes of leaves and sores, giving important structure information for plant disease detection. Edge recognition lets the model focus on the most important parts of the picture, like disease spots and damage areas, by finding sharp changes between different areas.

e. Image Resizing:

Resizing pictures is an important part of preprocessing that makes sure the input images for deep learning models are all the same size. Images of plant leaves can be different sizes, but for most machine learning models to work well, the input size needs to be constant. Setting all pictures to the same size makes sure they are all the same, which lets the model process groups of images in a way that can be scaled up or down. Figure 1 shows the steps used to prepare a picture for model training, such as improving it, shrinking it, and getting rid of noise.

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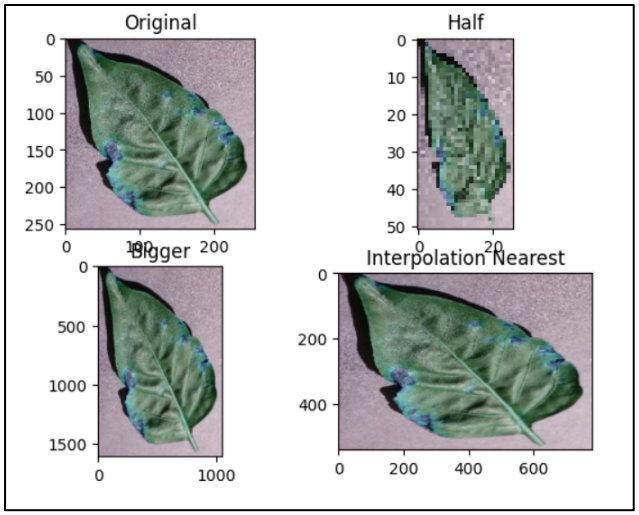

Figure 1: Dataset Pre-processing Image

C. Image Normalization

Image normalisation is an important step before processing. It changes the values of an image's pixels to a standard scale, usually between 0 and 1, or -1 and 1. This step makes sure that the model always gets data from the same range. This accelerates instruction and improves convergence during model optimisation. In the search for plant leaf diseases, normalisation helps reduce the impact of variations in illumination, camera settings, and leaf colour.

D. Noise Reduction

One of the most important steps in preparation is noise reduction, which eliminates random changes or undesired image defects that could compromise illness identification's accuracy. Noise in images of plant leaves might be from a camera malfunction, the surroundings, image reduction, or another source entirely. People may use median filtering, bidirectional filtering, or Wiener filtering to eliminate high-frequency noise and smooth out the image. By removing undesired artefacts and maintaining significant structural features like leaf veins and tumours, noise reduction helps to clarify the image. Eliminating noise helps the model to concentrate on the main trends connected to plant diseases. Figure 2 shows how noise reduction methods can be used to improve the quality of images for finding diseases. This makes disease recognition more accurate and reliable.

- Noise Model

$$\boldsymbol{I_{noisy(x,y)} = I_{clean(x,y)} + N(x,y)}$$

where I_noisy is the noisy image, I_clean is the original clean image, and N(x, y) represents noise.

- 2. Filter Function (e.g., Median Filter)

$$\boldsymbol{F(x,y) = median\left(I_{noisy(x+i,y+j)}\right) for\ i,j\ \in\ [-k,k]}$$

where F(x, y) is the filtered value at position (x, y), and k is the filter window size.

- 3. Apply Filtering to Remove Noise

$$\boldsymbol{I_{filtered(x,y)} = \Sigma[i = -k\ to\ k]\Sigma[j = -k\ to\ k]F(x+i,y+j) * W(i,j)}$$

where W(i, j) is the filter kernel (e.g., Gaussian or Sobel).

- 4. Resulting Denoised Image

$$\boldsymbol{I_{denoised} = I_{noisy} - N_{filtered}}$$

where N_filtered is the reduced noise after filtering.

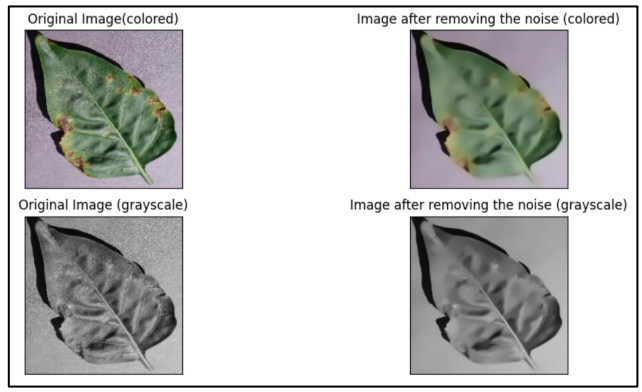

Figure 2: Noise Reduction Image

E. Data Augmentation

Randomly changing the pictures is a key part of data enrichment, which is a method for growing the variety and amount of training data. This is helpful when working with small datasets because it helps the model generalise better, which stops it from being too good at fitting the data and makes it work better on data it hasn't seen before. Left-right flipping, which involves flipping the picture horizontally, helps make mirror images of the original image. This keeps the model from being skewed by the way the plant leaves are arranged. This change helps the model learn what the leaf looks like no matter which way it is facing. One more usual enhancement is turning the picture 90 degrees. This makes the model more stable when the leaves' direction changes by simulating the different views and angles that the leaves might be seen from in real life. By "saturating" the picture, or making the colours brighter by a number (like 3), the lighting looks like it would in real life. The model doesn't become too sensitive to changes in colour because of this transformation. Instead, it can focus on the leaf's structure, like tumours or spots. If you want to practise spotting plant diseases at different sizes or distances, you can scale the picture. This makes sure that the model can deal with different leaf sizes without losing important data. In order to translate a picture, it needs to be moved around a bit within the frame.

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IV. Transfer Learning Algorithms

A. Fine-tune Lightweight MobileNetV2

Getting better MobileNetV2 involves making changes to a lightweight model that has already been trained to fit a certain job, like finding diseases on plant leaves. MobileNetV2 uses little memory and processing, which makes it perfect for use in places with limited resources, like mobile devices. The initial layers are frozen to adjust a model as they only capture broad characteristics. Retrained using the plant disease dataset, the remaining layers enables one to concentrate on identifying relevant patterns and illnesses. This procedure maintains the model's efficiency as well as feasible while increasing its accuracy.

Freeze Layers and Initialize New Layers

$$\boldsymbol{Model_{fine} - tuned = MobileNetV2_{pre} - trained + New\ Fully\ Connected\ Layers}$$

Freeze Initial Layers and Train Final Layers

$$\boldsymbol{L_{fine} - tuning = \left(\frac{1}{N}\right) * \Sigma[i = 1\ to\ N] Loss(y_{hat_i}, y_i)}$$

where y_hat_i = f(Model_fine-tuned(x_i))

Backpropagation and Update Weights

$$\boldsymbol{\theta_{updated} = \theta_{old} - \eta * \nabla\theta\ L}$$

where θ represents the model weights, η is the learning rate, and L is the loss function.

```
Total params: 2921551 (11.14 MB)
Trainable params: 663567 (2.53 MB)
Non-trainable params: 2257984 (8.61 MB)
```

B. Fine-tune Lightweight DenseNet121

Fine-tuning DenseNet121 involves modifying a DenseNet model previously taught to match a certain application, like as disease detection on plant leaves. DenseNet121 is well-known for its dense connection that is, for each layer receiving input from every layer below it. This facilitates the run of gradients and the reuse of features. When you fine-tune, you freeze the early layers catching simple lines and patterns that define fundamental characteristics. The last few layers are then retrained using the dataset for plant diseases so that they can focus on the current job. Fine-tuning DenseNet121 helps make the most of its strong feature extraction capabilities, which improves the accuracy of finding different plant diseases while keeping the computer's performance high.

Freeze Initial Layers and Initialize New Layers

$$\boldsymbol{Model_{fine} - tuned = DenseNet121_{pre} - trained + New\ Fully\ Connected\ Layers}$$

2. Forward Pass and Calculate Loss

$$\boldsymbol{L_{fine} - tuning = \left(\frac{1}{N}\right) * \Sigma[i = 1\ to\ N] Loss(y_{hat_i}, y_i)}$$

where y_hat_i = f(Model_fine-tuned(x_i))

3. Backpropagation to Calculate Gradients

$$\boldsymbol{\nabla\theta\ L = \frac{\partial L_{fine} - tuning}{\partial\theta}}$$

4. Update Weights Using Gradient Descent

$$\boldsymbol{\theta_{updated} = \theta_{old} - \eta * \nabla\theta\ L}$$

where θ represents the model weights, η is the learning rate, and L is the loss function.

```
Model: "sequential_3"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 densenet121 (Functional)    (None, 4, 4, 1024)        7037504

 flatten_4 (Flatten)         (None, 16384)             0

 dense_11 (Dense)            (None, 512)               8389120

 dropout_7 (Dropout)         (None, 512)               0

 dense_12 (Dense)            (None, 15)                7695

=================================================================
Total params: 15434319 (58.88 MB)
Trainable params: 8396815 (32.03 MB)
Non-trainable params: 7037504 (26.85 MB)
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V. Results and Discussion

A. Results of Lightweight MobileNetV2

Figure 3 shows the outcomes of the Lightweight MobileNetV2 model in finding diseases on plant leaves. MobileNetV2, which is known for being efficient and requiring less computing power, performs well on important measures like accuracy, precision, memory, and F1-score, all getting 94%. These findings reveal that the model can regularly classify plant diseases in the correct group. Though it is somewhat tiny and light, MobileNetV2 can extract significant characteristics from photos of plant leaves. This indicates that, given limited resources, such mobile devices, it is a useful tool for real-time disease detection in locations. Using a large scale in precision agriculture, the image in Figure 3 demonstrates how effectively the model can distinguish between healthy and ill plant leaves, therefore indicating its applicability. Figure 3 shows the remarkable speed and accuracy with which Lightweight MobileNetV2 finds illnesses.

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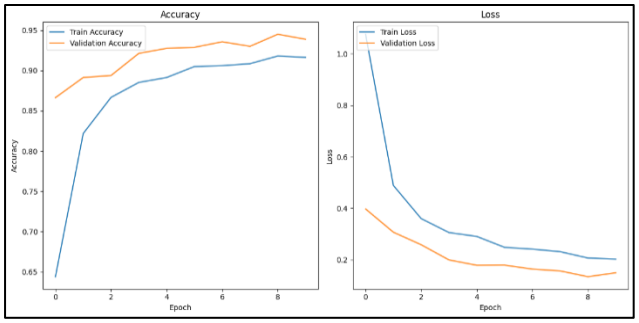

Figure 3: Results of Lightweight MobileNetV2

Based on this performance, MobileNetV2 might be a helpful tool for farmers and other professionals in the field seeking for automated and quick solutions to monitor illnesses in crops.

b. Results of Lightweight DenseNet121

Using the Lightweight DenseNet121 model to identify illnesses on plant leaves produces results shown in Figure 4. DenseNet121 performs quite well with a 90% accuracy, 90% precision, and 90% memory capacity. It is known for its dense connectivity and efficient feature reuse. Even though DenseNet121 uses more computing power than MobileNetV2, its deeper design and the way it joins layers make it great at picking out complex patterns in pictures of plant leaves. But these results show that it is good at finding different plant diseases, even though it did a little worse than MobileNetV2 in this dataset. Figure 4 shows a picture that shows how DenseNet121 can tell the difference between healthy and sick leaves.

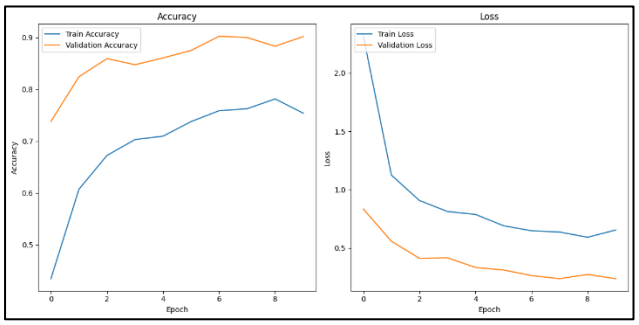

Figure 4: Results of Lightweight DenseNet121

This shows that the network could be useful in situations where a more complicated model is okay. In cases where more thorough feature extraction is needed, this model can be very helpful. It is also a good choice when computing power is not an issue.

C. Results of Fine-tune Lightweight Ensemble Transfer Learning Model

Figure 5 shows the outcomes of the Fine-tune Lightweight Ensemble Transfer Learning Model in finding diseases on plant leaves. This model takes the best parts of several pre-trained networks and uses them together to make the model work better overall. With an F1-score of 97%, it does better than both MobileNetV2 and DenseNet121 at finding and classifying plant diseases, with an accuracy of 97%, precision and recall at 96%, and an F1-score of 97%. The ensemble method improves generalisation by lowering the errors of each individual model. This makes forecasts more reliable.

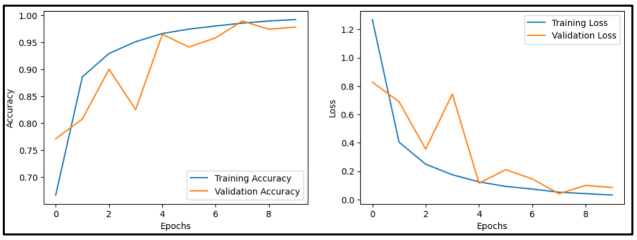

Figure 5: Results of Fine-tune Lightweight Ensemble Transfer Learning Model

This makes it a useful tool for finding diseases in agriculture in real time. The great results of this method show that ensemble learning can be useful for difficult tasks where accuracy is important for quick action and disease control.

D. Comparative Analysis of Models on Plant Leaf Diseases Dataset

TABLE 1: PERFORMANCE EVALUATION OF FINE-TUNED MODELS *FOR PLANT LEAF DISEASE DETECTION*

	Accuracy	**Precision**	**Recall**	**F1-Score**
Fine-tune Lightweight MobileNetV2	94	94	94	94
Fine-tune Lightweight DenseNet121	90	90	90	89
Fine-tune Lightweight Ensemble Transfer Learning Model	97	96	96	97

In Table 1, you can see a comparison of how well different fine-tuned models find plant leaf diseases. The Fine-tune Lightweight MobileNetV2, the Fine-tune Lightweight DenseNet121, and the Fine-tune Lightweight Ensemble Transfer Learning Model were all tested. MobileNetV2, a small model, got high scores in all categories—94% accuracy, precision, memory, and F1-score—showing that it is good at finding plant diseases. Even though DenseNet121 worked, it did a little worse than MobileNetV2 (90%) in accuracy, precision, and recall, and 89% in F1-score. Figure 6 shows how the different models performed, comparing their accuracy, precision, memory, and F1-score scores.

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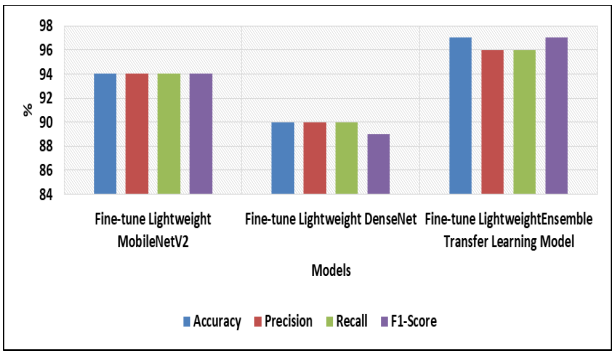

Figure 6: Comparative Analysis of Models on Plant Leaf Diseases Dataset

This suggests that the model might not be able to generalize as well for this job. The Ensemble Transfer Learning Model did the best, with 97% accuracy, 96% precision, 96% memory, and an F1-score. This shows that mixing different models can help improve recognition performance.

E. Comparative Analysis of Model Parameters

TABLE 2: COMPARATIVE ANALYSIS OF MODEL PARAMETERS *FOR PLANT LEAF DISEASE DETECTION*

Model	Total Param	Trainable Param	Non-trainable Param
Lightweight MobileNet	11.14 MB	2.53 MB	8.61 MB
Lightweight DesnsNet121	58.88 MB	32.3 MB	26.85 MB
Fine-tune Lightweight Ensemble Transfer Learning Model	76.74 MB	46.73 MB	30.01 MB

Table 2 shows how the model factors for three different designs used to find plant leaf diseases compare to each other. There are a total of 11.14 MB of parameters in the Lightweight MobileNet model, with 2.53 MB that can be trained and 8.61 MB that can't. This means that MobileNet is a small model that is designed to be efficient and can be used in places with limited resources. Lightweight DenseNet121, on the other hand, has 58.88 MB of parameters, of which 32.3 MB can be trained and 26.85 MB can't. Figure 7 shows the total, trainable, and non-trainable parameter sizes for each model and compares their parameters.

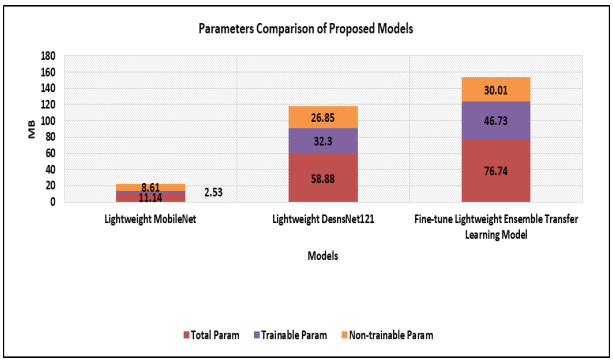

Figure 7: Comparative Analysis of Model Parameters

Although the ensemble model needs more computing power, the fact that it has more factors says that it is more complicated and might be more accurate.

VI. CONCLUSION

This research shows that deep learning methods, especially transfer learning and ensemble models, can be used to find diseases on plant leaves. Using pre-trained models like MobileNetV2 and DenseNet121 and fine-tuning makes it possible to classify things quickly and correctly even when there isn't a lot of labeled data. Using ensemble models improves performance even more by mixing the best features of several models. This leads to better accuracy, precision, recall, and F1-score. The outcomes reveal that MobileNetV2, even though it is small, works very well, while DenseNet121 gives solid outcomes with more parameters. The ensemble transfer learning model does the best, showing that it can be used for difficult jobs like finding plant diseases. This method not only gives a correct answer, but it can also be used on a large scale and is very reliable. This makes it an interesting tool for watching diseases in real time in precision agriculture. When these methods are used together, they could make it much easier to find diseases early and take care of crops, which would eventually increase farming output.

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